TUMOR LOCALIZATION IN BREAST PHANTOMS

CNN vs MLP

A matched, head-to-head comparison of two machine-learning approaches for microwave tumor localization

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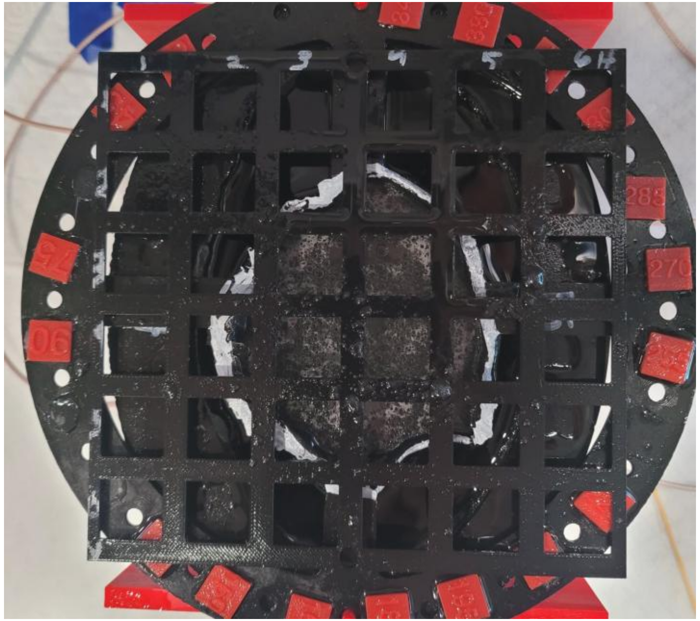
Preprocessing pipeline v1 (includes session-mean subtraction)

THE PROBLEM

Find a hidden tumor inside a breast phantom

- Goal: locate a small tumor surrogate (metal fishing weight) placed inside a realistic breast phantom, without opening it up.
- Microwave imaging: low-cost, non-ionizing alternative screening modality.
- The tumor perturbs how microwaves reflect and transmit through the tissue, but the change is tiny and buried in drift and noise.
- Worst case: a glandular insert (dense tissue) further masks the tumor signal.
- **Task: given one measurement, which of ~33–51 grid positions holds the tumor?**



A3 breast phantom with F5 glandular insert and measurement grid

THE SOLUTION

Let a machine learn the fingerprint of every position

Classic inverse-scattering reconstruction is slow and fragile. Instead, we treat localization as a pattern-recognition problem: measure the phantom with the tumor at every known position, then train a model to recognise each position's electromagnetic fingerprint.

16

S-parameters per sweep

791

frequency points, 0.1–8 GHz

~2–3%

chance-level accuracy

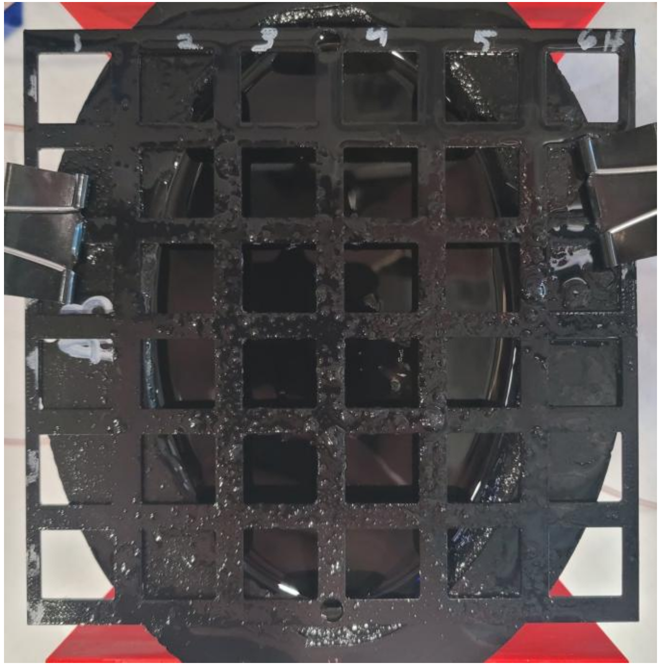
PART 1

Methods

The measurement, the two models, and how they differ

METHODS

The physical measurement



Empty A3 phantom: adipose bowl, 6×6 in grid, 4 antennas at the rim

- 4-port Hunter VNA sweeps 0.1–8 GHz (791 points) through 4 antennas around the phantom.
- Each sweep captures all 16 S-parameters: 4 reflections (S11…S44) plus transmissions between every antenna pair.
- Tumor surrogate is moved through a grid of cell corner positions; 16 repeat trials per position.
- Baseline (empty-phantom) sweeps recorded each session for calibration.
- **Each setup measured in 3–4 independent sessions on different setups/times; this matters later (LOSO).**

METHODS

Two ways to turn a measurement into a position

Both models receive the SAME preprocessed measurement and output a probability for every grid position. They differ in how they look at the data.

CNN

Convolutional Neural Network · MATLAB

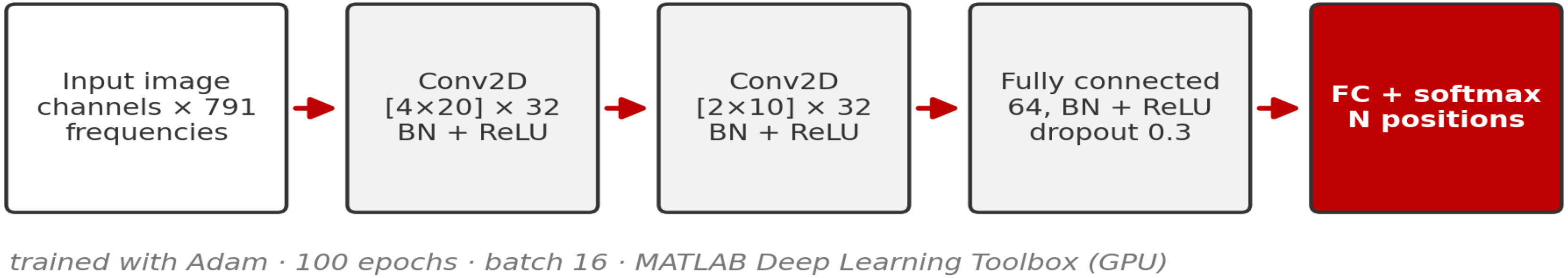
- Sees the measurement as a 2-D image: y axis = channels, x axis = frequency
- Convolution filters slide along frequency, learning local patterns

MLP

Multi-Layer Perceptron · Python

- Sees the measurement as one long flattened vector of numbers
- Every input connects to every neuron: the network has no built-in idea that neighboring frequencies are related, so any such structure must be learned from scratch

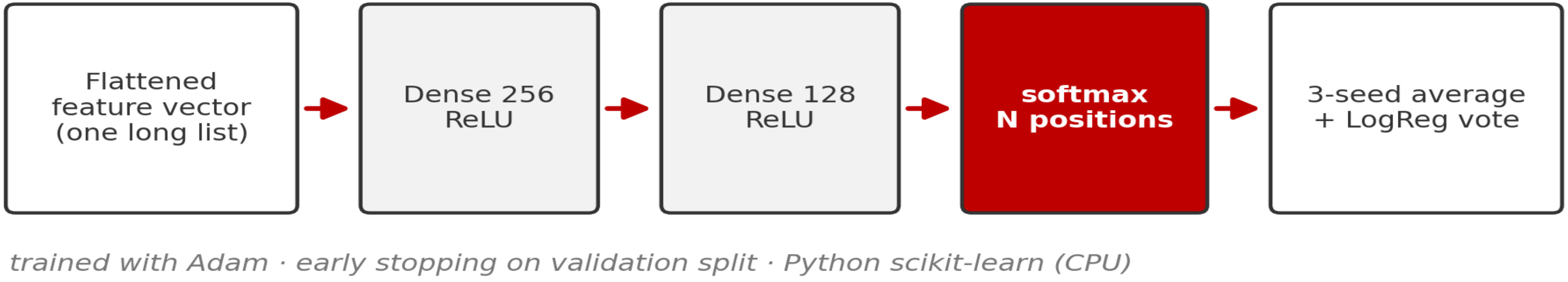
The CNN



- Input is an image: one row per data channel (e.g. |S11|, ∠S11, …), one column per frequency point (791).
- Two convolution layers (32 filters each) detect local patterns such as resonance shifts and ripples anywhere along the spectrum, sharing weights across frequency.
- Batch-normalisation + ReLU after every layer; 30% dropout fights overfitting.
- One network, trained 100 epochs with Adam (batch 16) on the GPU; softmax outputs one probability per grid position.

METHODS

The MLP



- Input is a flat vector: all channels and frequencies concatenated (thousands of features). No spatial structure is assumed.
- Two dense layers (256 → 128, ReLU): every feature can interact with every other, but nothing is shared or localized.
- Early stopping on a held-out validation split prevents overtraining.
- Ensemble: 3 networks with different random seeds are averaged, plus a logistic-regression model; the ensemble vote is the prediction.

CNN vs MLP: structural differences

	CNN	MLP
How it sees data	2-D image (channels × frequency)	1-D flattened vector
Inductive bias	Local patterns, shared across frequency	None; fully general
Parameters	Small conv filters + FC (~0.5–2 M)	Huge first layer (features × 256)
Training	Fixed 100 epochs, single network	Early stopping, 3-seed ensemble + LogReg
Framework	MATLAB Deep Learning Toolbox (GPU)	Python scikit-learn (CPU)

Same job, opposite philosophies: the CNN assumes frequency-local structure; the MLP assumes nothing.
Everything else in this study is held identical.

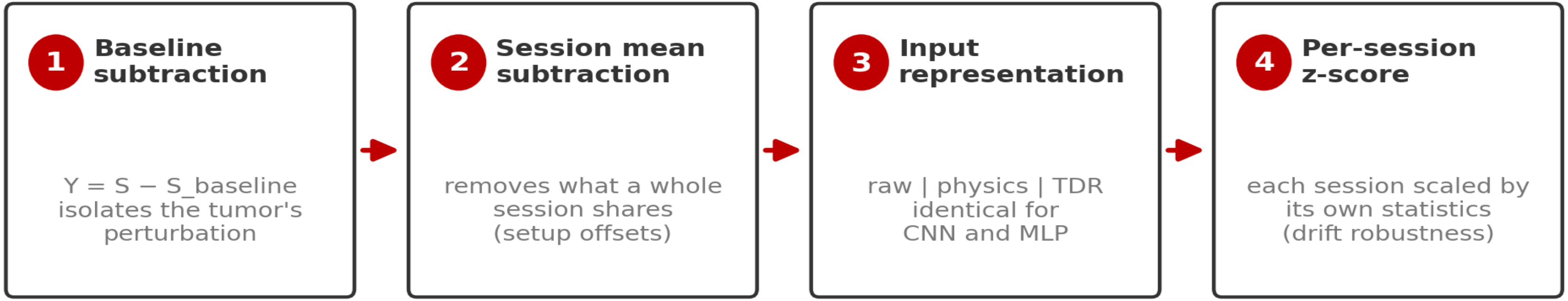
PART 2

The data pipeline

What happens to a measurement before any 'AI' ever sees it. Identical for both models.

DATA PIPELINE

Four steps from raw sweep to model input



- Steps 1–2 remove everything that is not the tumor; step 4 (per-session z-score) is the key drift defense: each session is standardized by its own statistics. Applied identically to CNN and MLP.
- **Upgrading the CNN to this pipeline was worth +17 pts on F5 single-antenna (59→76%) with fold-to-fold σ cut from 40% to 9%, and +15 pts on F4 single-antenna (82→97%). Its catastrophic cross-session failures disappeared entirely.**

DATA PIPELINE

Three input representations

RAW

Magnitude + phase of each calibrated S-parameter.

The measurement as the VNA reports it. Used for the main results.

PHYSICS

Adds real & imaginary parts and the IFFT time-domain (TDR) envelope.

Derived at no cost from the same data; covered at the end.

TDR-ONLY

Just the time-domain envelope, a pulse-echo view.

An ablation: how much lives in time domain alone?

Same numbers, two shapes:

CNN: channels stacked as image rows × 791 frequency columns
MLP: the same rows laid end-to-end as one long vector

Antenna subsets work the same way: '2 antennas' keeps only S-parameters among ports 1 & 3; '1 antenna' keeps only S11.

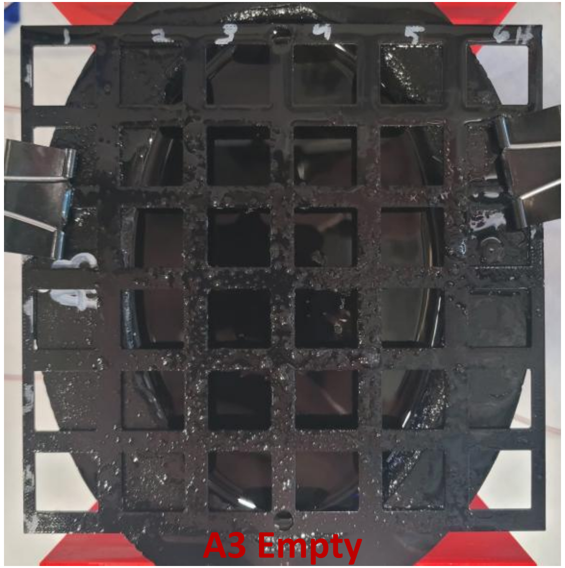
DATA PIPELINE

How we score: Leave-One-Session-Out

Leave-One-Session-Out: the model NEVER sees the test session, so the score reflects a brand-new day of measurements

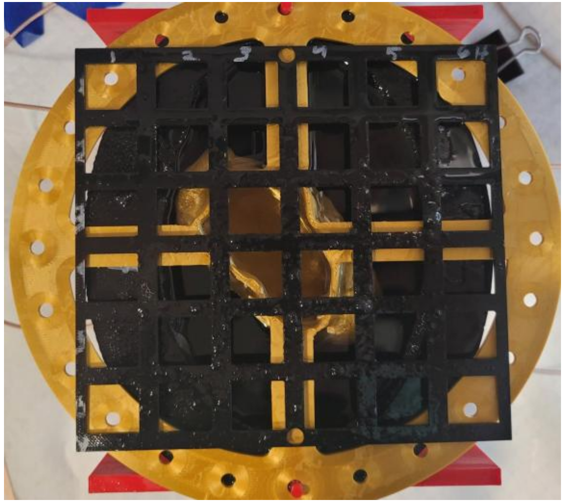
- A random 75/25 split mixes trials from the same session into train AND test, so the model can cheat off session drift, inflating accuracy.
- LOSO trains on N−1 sessions and tests on the session it has never seen; every session takes a turn. We report per-position majority vote over its 16 trials.
- **We have thoroughly stress-tested LOSO and verified that it works: under LOSO, models cannot exploit session drift, and drift-only information scores near chance.**

Three phantom setups, increasing difficulty



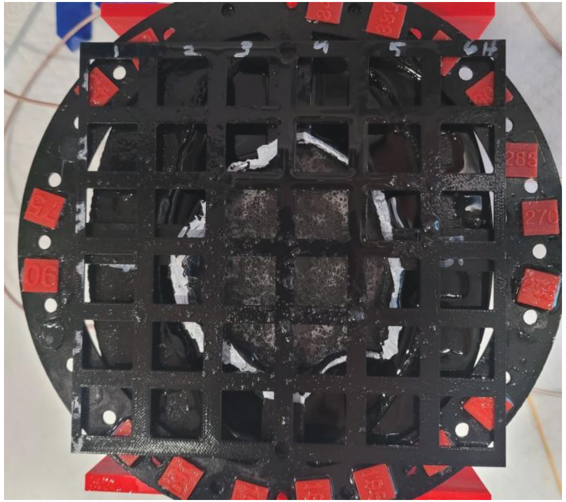
A3 Empty

adipose only
51 positions · 3 sessions



A3 + F4

small glandular insert
39 positions · 4 sessions



A3 + F5

large glandular insert
33 positions · 3 sessions (latest)

PART 3

Results

Per-position vote accuracy (%) under LOSO · identical raw inputs (mag + phase) · chance = 2–3%

RESULTS · RAW INPUTS

All 4 antennas: full S-parameters (16)

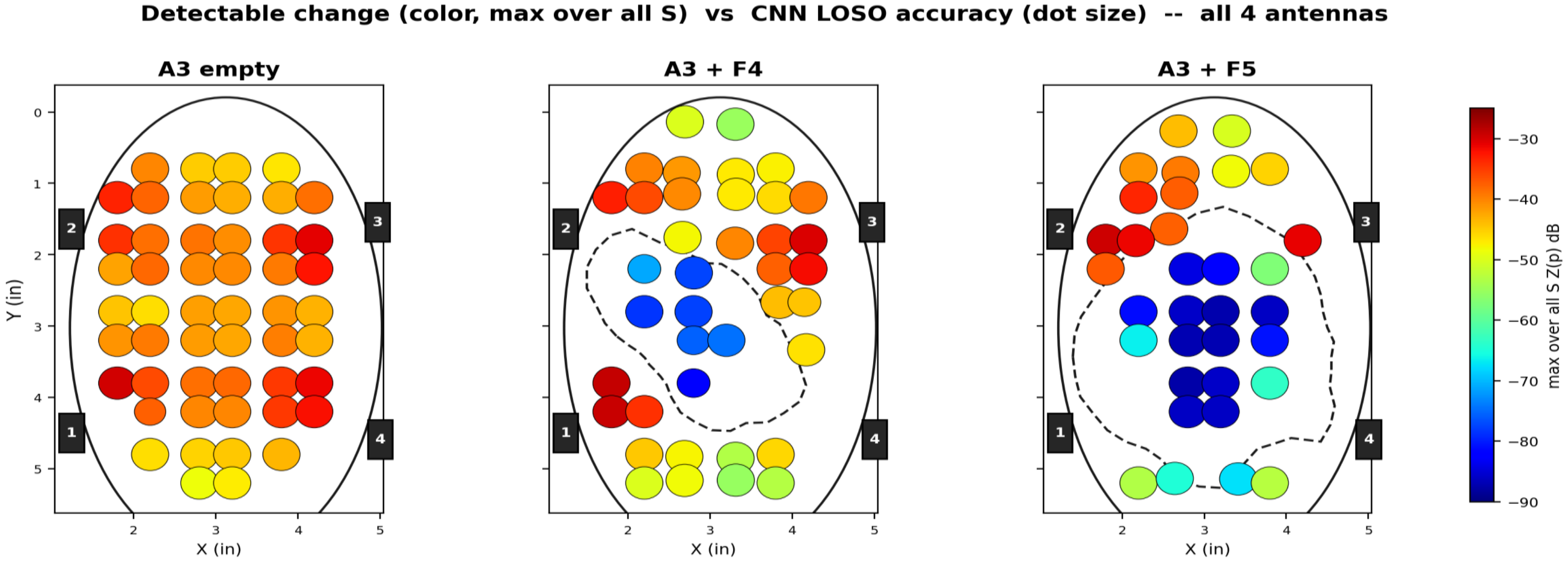
Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	**99.3 ± 1.1**	99.3 ± 0.9	**+0.0**
A3 + F4	39-way	**97.4 ± 3.0**	97.4 ± 1.8	**+0.0**
A3 + F5	33-way	**100.0 ± 0.0**	96.0 ± 2.9	**+4.0**

Mean ± std over LOSO folds. Bold = better method.

With the full antenna array, both methods are near-perfect on every phantom; drift and glandular masking are fully compensated. CNN edges ahead on the hardest setup (F5).

Error Locations: all 4 antennas

Dot color = how strongly the tumor physically changes the signal (detectable difference). **Dot size** = CNN LOSO accuracy at that position. Dashed outline = glandular insert.



Detectable change still dips inside the glandular insert (blue), but with 16 complementary views the classifier compensates: accuracy dots stay large almost everywhere. MLP versions look qualitatively identical.

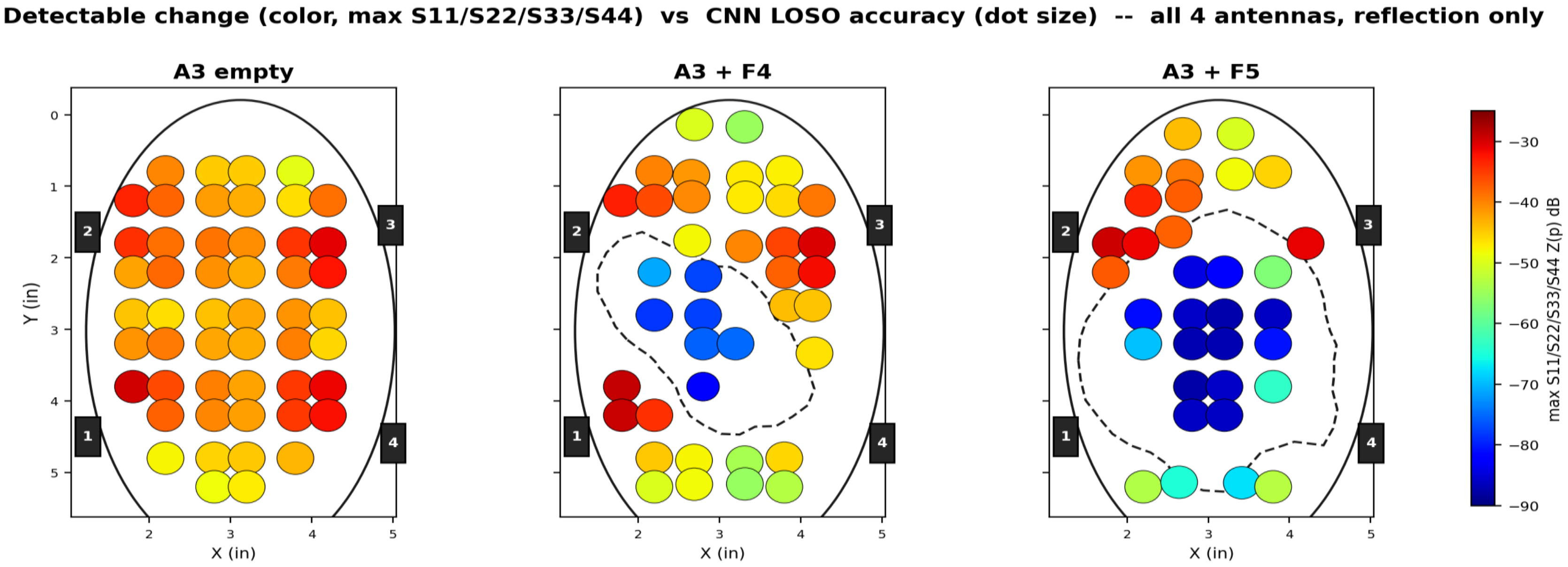
All 4 antennas: reflection only (S11, S22, S33, S44)

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	**100.0 ± 0.0**	99.3 ± 0.9	**+0.7**
A3 + F4	39-way	**98.7 ± 1.5**	93.6 ± 4.3	**+5.1**
A3 + F5	33-way	**100.0 ± 0.0**	88.9 ± 2.9	**+11.1**

No through-phantom transmission paths; each antenna only listens to its own echo.

Dropping transmission barely touches the CNN (F5 stays at 100%), while the MLP gives up 4-7 points on the glandular phantoms. Four reflection coefficients are enough for a CNN.

Error Locations: all 4 antennas, reflection only



Reflection-only detectable change (max of S11, S22, S33, S44). The insert still carves a low-signal pocket, but CNN accuracy dots stay large.

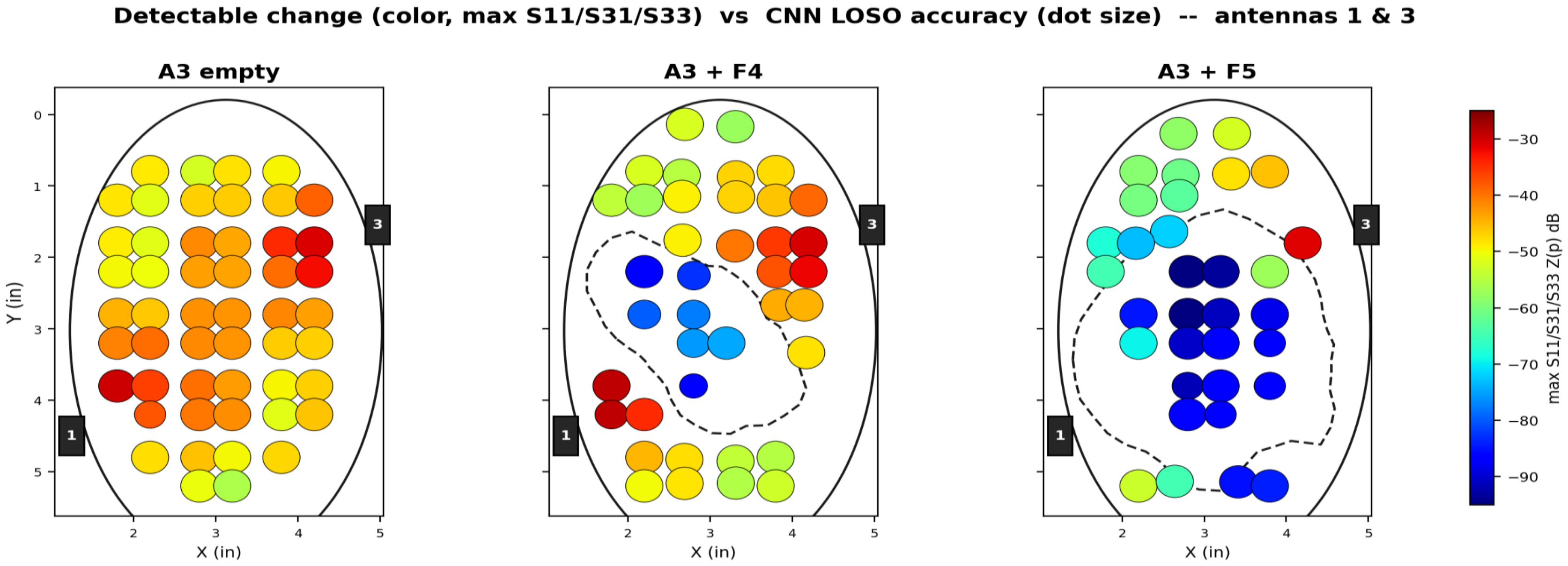
2 antennas (ports 1 & 3): full S-parameters

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	**99.3 ± 1.1**	99.3 ± 0.9	**+0.0**
A3 + F4	39-way	**95.5 ± 1.3**	93.6 ± 4.3	**+1.9**
A3 + F5	33-way	**96.0 ± 7.0**	81.8 ± 4.3	**+14.2**

Antennas 1 & 3 sit on opposite sides of the phantom (most informative pair).

Halving the antennas separates the methods: the CNN stays high (96%+ everywhere) while the MLP drops 14 points on F5. Convolutional structure extracts more from less hardware.

Error Locations: 2 antennas (1 & 3)



With two antennas a low-signal pocket appears inside the glandular insert, and accuracy dots shrink with it.

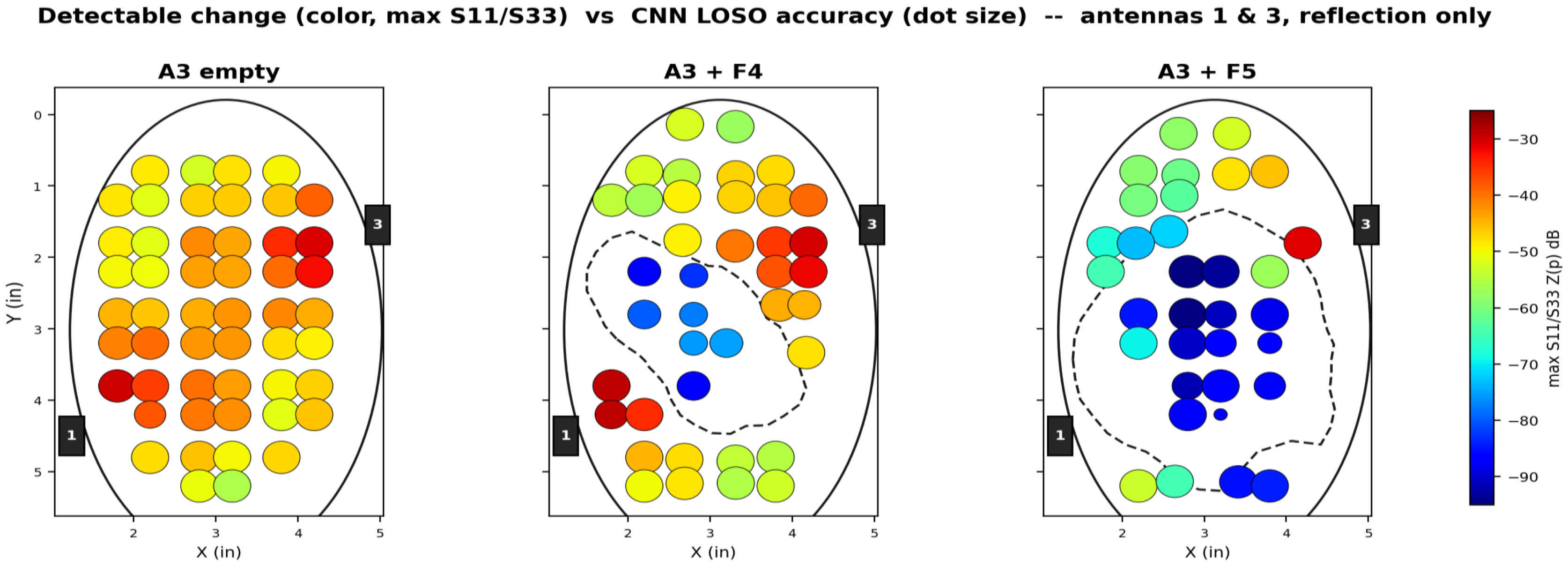
2 antennas (1 & 3): reflection only (S11, S33)

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	**99.3 ± 1.1**	99.3 ± 0.9	**+0.0**
A3 + F4	39-way	92.3 ± 5.9	**93.6 ± 5.9**	**−1.3**
A3 + F5	33-way	**90.9 ± 5.3**	74.8 ± 6.2	**+16.1**

The toughest 2-antenna condition: two reflection coefficients and nothing else.

The CNN holds 91-99% while the MLP falls to 75% on F5. F4 is the single config in the study where the MLP edges the CNN (93.6 vs 92.3).

Error Locations: 2 antennas (1 & 3), reflection only



With only S11 and S33, the low-signal pocket deepens and CNN errors start to appear inside the insert.

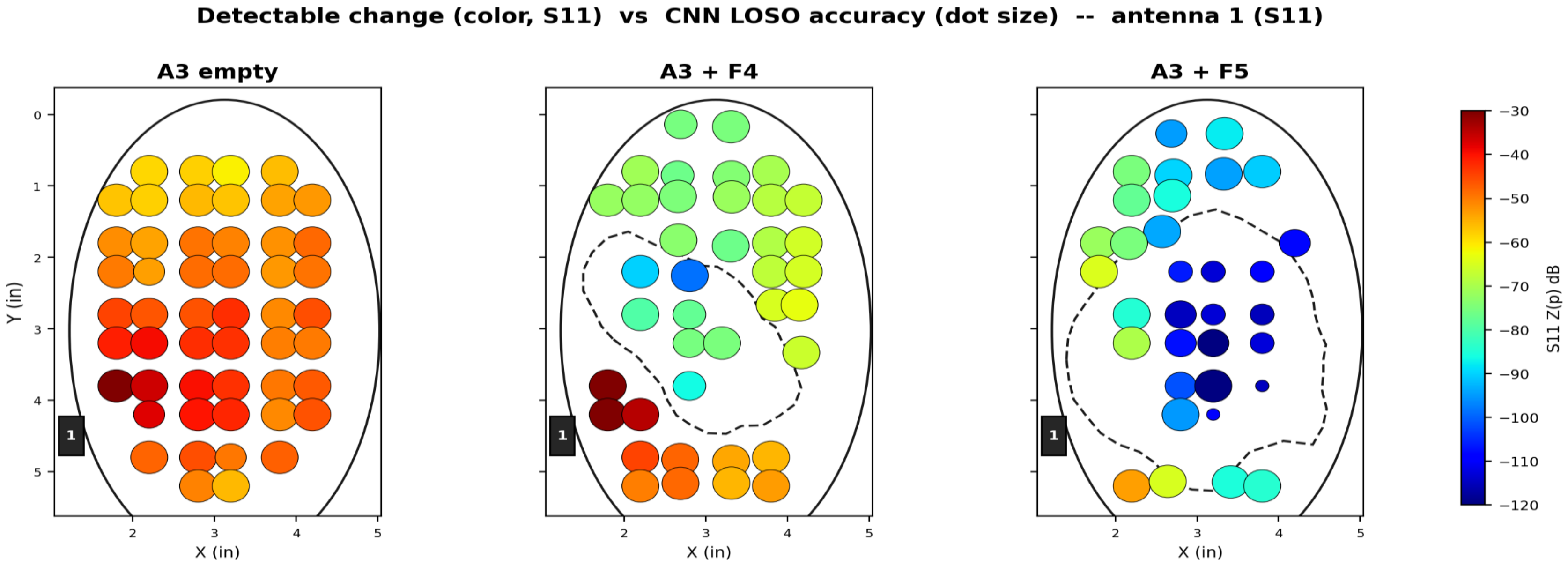
1 antenna: S11 only

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	98.0 ± 2.0	**99.3 ± 0.9**	**−1.3**
A3 + F4	39-way	**96.8 ± 4.9**	91.0 ± 5.6	**+5.8**
A3 + F5	33-way	**75.8 ± 9.1**	58.6 ± 9.4	**+17.2**

A single reflection coefficient: the minimal possible hardware.

The extreme case: one antenna, one S-parameter. The CNN holds 76–98%; the MLP falls to 59% on F5. The gap grows exactly as the task gets harder, and the F5 errors cluster inside the glandular insert (next slide).

Error Locations: 1 antenna (S11)



Single antenna (S11): small blue dots cluster inside the insert. Where physics buries the tumor, the classifier fails.

PART 4

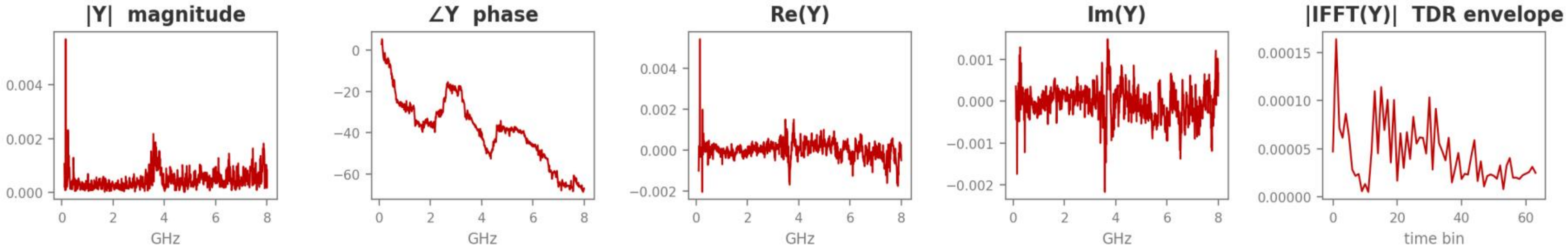
Physics-informed features

Squeezing more out of the same measurement: no new hardware, no new data collection

PHYSICS-INFORMED FEATURES

One complex measurement, five physical views

Each calibrated S-parameter Y(f) is one complex spectrum. From it we derive, at zero measurement cost: magnitude & phase (resonance behaviour), real & imaginary parts (full complex information), and the inverse-FFT envelope, a time-domain reflectometry (TDR) view where the tumor appears as an echo at its round-trip delay.



Real example: A3+F5 phantom, S11, tumor at grid position R3C4 (calibrated response Y)

The 'physics' input feeds all five views to both models: the CNN as extra image rows, the MLP as extra vector features. Identical information, matched again.

Same tests, physics inputs

All-4-antenna physics: both models at ceiling, 97–100% (F5 reaches 100 / 100). Below: the configs where it matters. Reflection-only: next slide.

2 antennas (1 & 3), physics input

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	**99.3 ± 1.1**	99.3 ± 0.9	**+0.0**
A3 + F4	39-way	**98.1 ± 1.3**	92.3 ± 3.1	**+5.8**
A3 + F5	33-way	**98.0 ± 3.5**	85.9 ± 7.6	**+12.1**

1 antenna (S11), physics input

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	**98.7 ± 1.1**	98.7 ± 1.8	**+0.0**
A3 + F4	39-way	**95.5 ± 4.4**	91.7 ± 5.8	**+3.8**
A3 + F5	33-way	**79.8 ± 3.5**	62.6 ± 8.0	**+17.2**

Reflection only + physics features

All 4 antennas, reflection only (S11, S22, S33, S44), physics input

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	**100.0 ± 0.0**	99.3 ± 0.9	**+0.7**
A3 + F4	39-way	**98.1 ± 1.3**	94.2 ± 3.3	**+3.9**
A3 + F5	33-way	**100.0 ± 0.0**	87.9 ± 6.6	**+12.1**

2 antennas (1 & 3), reflection only (S11, S33), physics input

Phantom setup	Positions	CNN (%)	MLP (%)	CNN – MLP
A3 Empty	51-way	**100.0 ± 0.0**	98.7 ± 1.9	**+1.3**
A3 + F4	39-way	**94.2 ± 3.2**	90.4 ± 3.3	**+3.8**
A3 + F5	33-way	**97.0 ± 5.3**	79.8 ± 7.6	**+17.2**

With physics features, reflection-only costs the CNN almost nothing (F5: 100% with four antennas, 97% with two); the MLP pays 8-12 points. A reflection-only system is practical with a CNN.

What the derived features buy, and what TDR alone can't

Physics vs raw (percentage points gained)

- F5, 2 antennas: CNN +2.0 (96→98) · MLP +4.1 (82→86)
- F5, 1 antenna: CNN +4.0 (76→80) · MLP +4.0 (59→63)
- F5, all 4: MLP +4.0 (96→100), joining CNN at 100%
- Easy setups (empty, F4 full-array): no change, already at ceiling.
- **Physics features help most exactly where raw inputs struggle.**

TDR-only ablation (selected)

- F4, 1 antenna: CNN 84 / MLP 64 (vs 97 / 91 raw): a big loss for both.
- But the CNN mines far more from TDR alone (F5 all-4: 98 vs MLP's 76).

Additional finding: hierarchical (quadrant → position) classification

- Two-stage 'find the corner first' scheme tested on both models.
- It never helps, and hurts the hardest cases: CNN −6 on F5, MLP −6 on F4-single.
- **Wrong quadrant = unrecoverable. Single-stage wins under LOSO.**

CONCLUSIONS

What we learned

CNN ≥ MLP, everywhere

On matched inputs the CNN ties easy cases and wins hard ones by up to 17 points.

The gap grows with difficulty

Fewer antennas + bigger glandular insert = bigger CNN advantage.

Preprocessing drives robustness

Per-session z-score eliminated the CNN's catastrophic cross-session failures.

Physics features add accuracy at no cost

Derived views (+TDR) add up to 4 points on the hardest configurations.

Errors follow the physics

Misses cluster inside the glandular insert, where detectable signal change collapses.

Fewer antennas is viable with a CNN

2 antennas: 96–98% on every phantom. Even 1 antenna: 76–98%.